Marmara University
Electrical and Electronincs Engineering
Probability and Statistics
Problem Sets with Solutions #2

[1] Prove that if $x \in A \cap B^c$, then $x \in A-B$.

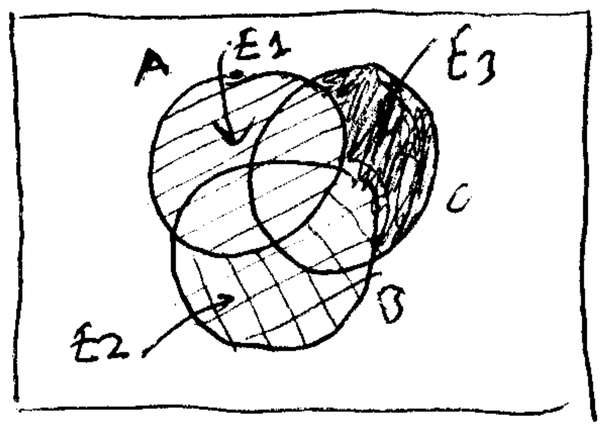
SOLUTION: If $x \in A \cap B^c$, then by definition $x \in A$ and $x \in B^c$.
But, if $x \in B^c$, then $x \notin B$. Thus, $x \in A$ and $x \notin B$.
By definition, $x \in A-B$ and $A-B^c \subset A-B$. ∎

[2] In this problem, we see how to construct disjoint sets from ones that are not disjoint so that their unions will be the same. We consider only three sets. Calling the nondisjoint sets A, B, C and the union $D = A \cup B \cup C$, we wish to find three disjoint sets E_1, E_2 and E_3 so that $D = E_1 \cup E_2 \cup E_3$. To do so, let

$$E_1 = A$$
$$E_2 = B - E_1$$
$$E_3 = C - (E_1 \cup E_2)$$

Using a Venn diagram explain this procedure. If we now have sets $A_1, A_2, \ldots, A_N$ explain how to construct N disjoint sets with the same union.

SOLUTION:



$E_1 = A$
$E_2 = B - E_1$
$E_3 = C - (E_1 \cup E_2)$

In general,

$$E_1 = A_1$$
$$E_2 = A_2 - E_1$$
$$E_3 = A_3 - (E_1 \cup E_2)$$
$$E_4 = A_4 - (E_1 \cup E_2 \cup E_3)$$
$$\vdots$$
$$E_N = A_N - (E_1 \cup E_2 \cup \ldots \cup E_{N-1}) = A_N - \bigcup_{k=1}^{N-1} E_i$$

∎

[3] Replace the set expression $A \cup B \cup C$ with one using intersections and complements. Replace the set expression $A \cap B \cap C$ with one using unions and complements.

SOLUTION:

$$A \cup B \cup C = (A^c \cap B^c \cap C^c)^c$$
$$A \cap B \cap C = (A^c \cup B^c \cup C^c)^c$$

∎

[4] Prove that if the sample space size has size N, the total numbe of events (including the impossible event and the certain event) is 2^N. (Hint: There are $\binom{N}{k}$ ways to choose an event with k outcomes from a total of N outcomes. Also, use the binomial formula $(a+b)^N = \sum_{k=0}^{N} \binom{N}{k} a^k b^{N-k}$)

Solution The numbe of events with 0 outcomes is $\binom{N}{0}$
" " " " " 1 " is $\binom{N}{1}$

The numbe of events with N outcomes is $\binom{N}{N}$

The total numbe of events: $\binom{N}{0}+\binom{N}{1}+\cdots+\binom{N}{N} = \sum_{k=0}^{N}\binom{N}{k} = \sum_{k=0}^{N}\binom{N}{k} 1^k \cdot 1^{N-k}$

$= (1+1)^N = 2^N$ //

[5] Prove that $P[\bigcup_{i=1}^{N} E_i] = \sum_{i=1}^{N} P[E_i]$ for all E_i's mutually exclusive

Use mathematical induction

SOLUTION. ~~Assume~~ ~~that~~ The result is true for $N=2$ $P[E_1 \cup E_2] = P[E_1] + P[E_2]$

Assume that it is true for N. Then, for $N+1$,

$$P[\bigcup_{i=1}^{N+1} E_i] = P[\{\bigcup_{i=1}^{N} E_i\} \cup E_{n+1}]$$

$$= P[\bigcup_{i=1}^{N} E_i] + P[E_{n+1}] \quad \text{.. } (E_i\text{'s are all disjoint})$$

$$= \left(\sum_{i=1}^{N} P[E_i]\right) + P[E_{n+1}]$$

$$= \sum_{i=1}^{N+1} P[E_i]$$

In other words, the result is true for $N+1$. Hence it must be true for all N.

[6] A numbe, is chosen from $\{1,2,3,\ldots\}$ with probability

$$P[i] = \begin{cases} \frac{4}{7}, & i=1 \\ \frac{2}{7}, & i=2 \\ (\frac{1}{8})^{i-2}, & i \geq 3 \end{cases}$$

Find $P[i \geq 4]$.

SOLUTION: $P[i \geq 4] = 1 - P[i<4] = 1 - [P[i=1] + P[i=2] + P[i=3]]$

$= 1 - [\frac{4}{7} + \frac{2}{7} + \frac{1}{8}]$

$= \frac{1}{56}$

[7] Determine the probability that in a class of 23 students two or more students have birthdays on January 1.

SOLUTION: $P[2 \text{ or more}] = 1 - [P[0] + P[1]]$

$$= 1 - \left[\frac{364^{23}}{365^{23}} + \frac{23 \cdot 364^{22}}{365^{23}}\right] = 0.0018$$

□

[8] A pizza can be ordered with up to four different toppings. Find the total number of different pizzas (including no toppings) that can be ordered. If a person wishes to pay for only two toppings, how many two-topping pizzas can he order?

SOLUTION: Method 1:

$\binom{4}{0}$ no toppings
$\binom{4}{1}$ 1 topping
$\binom{4}{2}$ 2 toppings
$\binom{4}{3}$ 3 toppings
$\binom{4}{4}$ 4 toppings

Total number $\sum_{k=0}^{4} \binom{4}{k} = 2^4 = 16$.

Method 2: Each pizza ordered can be represented as 4-tuple with elements 0 and 1. Only toppings 1 and 4 can be denoted as, for example

$(1, 0, 0, 1)$

Hence, there are $2^4 = 16$ possible 4-tuples. □

[9] List all the combinations of two coins that can be chosen from the following coins: one penny (1 cent), one nickel (5 cents), one dime (10 cents) and one quarter (25 cents). What are the possible sum values?

SOLUTION: There are a total of $\binom{4}{2} = 6$ combinations. They are listed below with their corresponding sum values.

Combination	Sum Value
p, n	6
p, d	11
p, q	26
n, d	15
n, q	30
d, q	35

p: penny
n: nickel
d: dime
q: quarter

[10] A class consists of 30 students, of which 20 are freshmen and 10 are sophomores If 5 students are selected at random, what is the probability that they will all be sophomores

SOLUTION: Size of the sample space $\binom{30}{5}$

The numbe of ways 5 students can be selected from 10 sophomores $\binom{10}{5}$

The number of ways 0 students can be selected from 20 freshme $\binom{20}{0}$

$$\text{Hence } P[\text{event}] = \frac{\binom{10}{5}\cdot\binom{20}{0}}{\binom{30}{5}} = \frac{(10!)/(5!)^2}{30!/(25!\,5!)}$$

$$= \frac{25! \times 10!}{30! \times 5!} = \frac{10\cdot 9\cdot 8\cdot 7\cdot 6}{30\cdot 29\cdot 28\cdot 27\cdot 26} = 0.0018 \quad \boxtimes$$

[11] An urn contains an infinite numbe of coins that are eithe two-headed or two-tailed The proportion of each kind is the same If we choos M coins at random, explain why the probabilit of obtaining k heads is given by $\binom{M}{k} p^k (1-p)^{M-k}$ with $p = 1/2$. Also, how does this experiment compare to the tossing of a fair coin M times?

SOLUTION: Each draw produces a head or tail with

$$P[\text{head}] = P[\text{tail}] = \frac{1}{2}$$

Since the proportions are equal This is the same as of the urn containing an infinite numbe of red balls (heads) and black balls (tails) Each outcome can be represented as an M-tuple of H's and T's (or equivalently of R's and B's) For example,

$$\underbrace{(H, H, T, \ldots, H)}_{M\text{-terms}}$$

If each outcome of M successive coin tosses is equally likely, then this experiment is equivalent to choosing M balls at random.